Single agent xN vs MAPPO vs IPPO: robots in the obstacle map, collisions end the episode

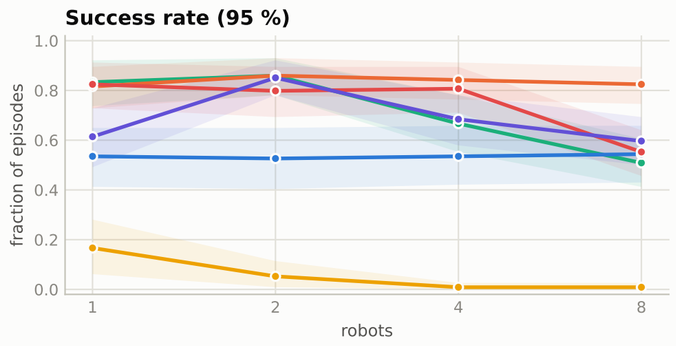
Same episodes for every policy (maps, start poses, seed), deterministic actions. Lines: mean over maps, band: 95 % bootstrap CI over maps.
Without flip_when_stuck. Team efficiency relative to Single-agent xN with 1 robot on the same map. Time to the goal: only successful episodes. Every method: lidar detections of robots in the obstacle map, the first collision ends the episode.

Single-agent xN · Single-robot independent xN · IPPO · MAPPO · IPPO retrained · MAPPO retrained

paper

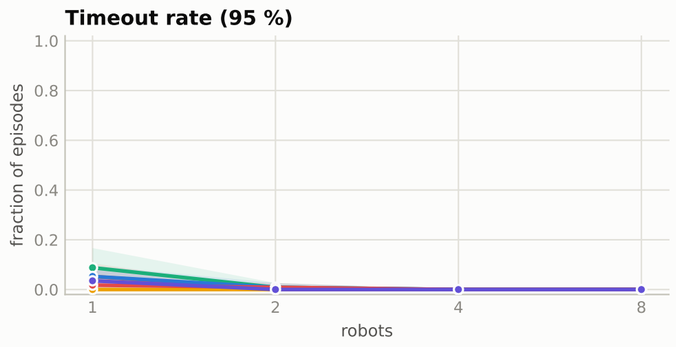
38 maps x 3 episodes, goal 95 % coverage, limit 333 min

Success rate (95 %)

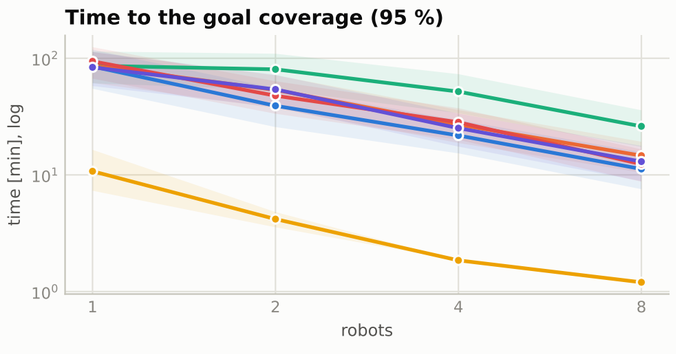


Collision rate

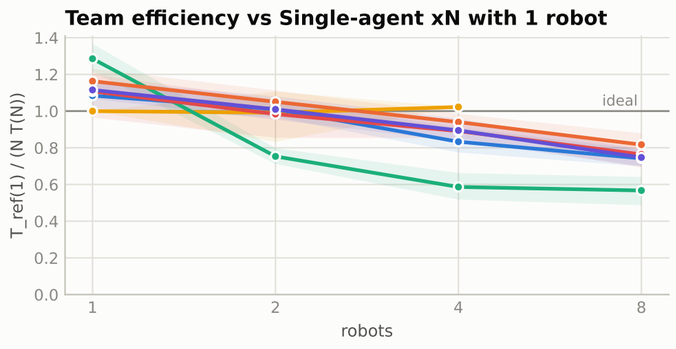
Timeout rate (95 %)



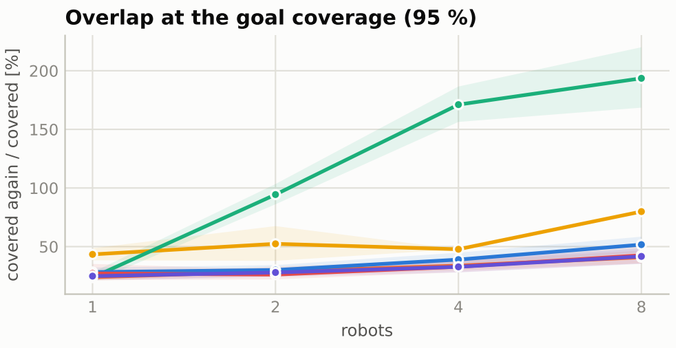
Time to the goal coverage (95 %)



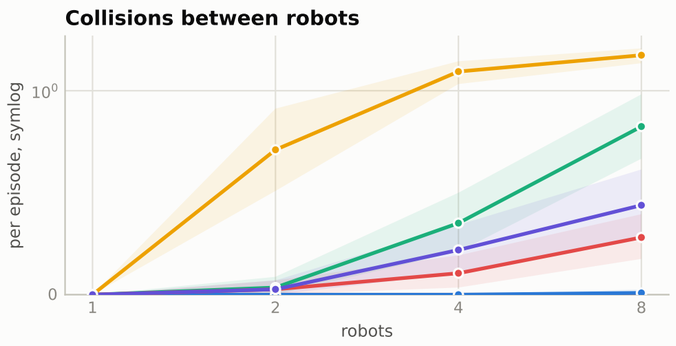
Team efficiency vs Single-agent xN with 1 robot



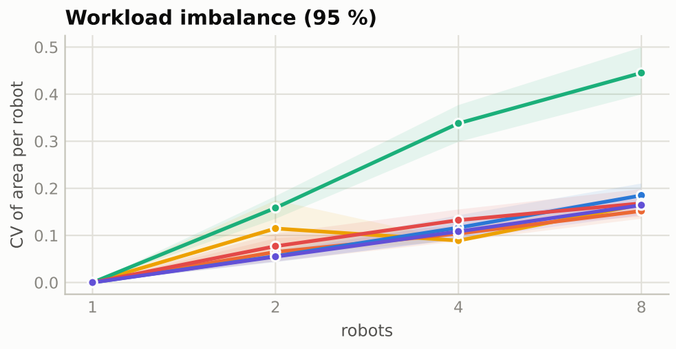
Overlap at the goal coverage (95 %)



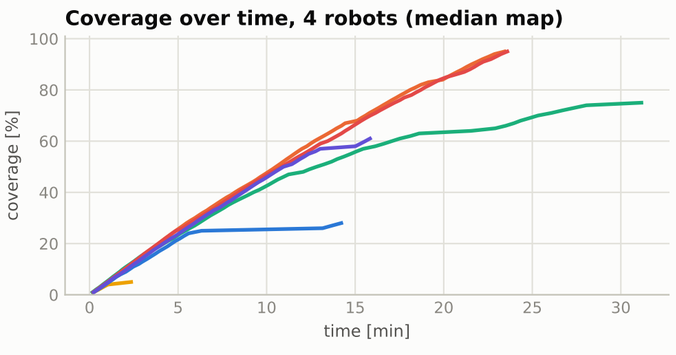
Collisions between robots



Workload imbalance (95 %)



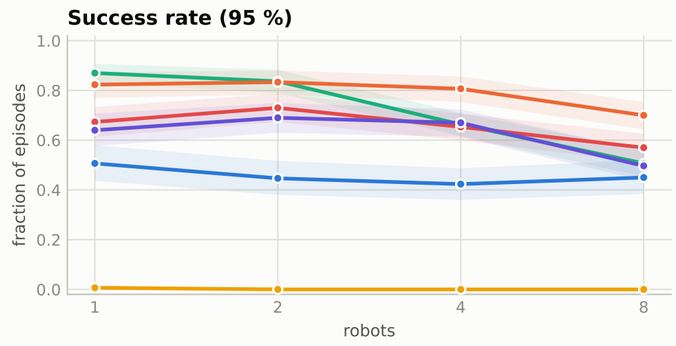
Coverage over time, 4 robots (median map)



test_large

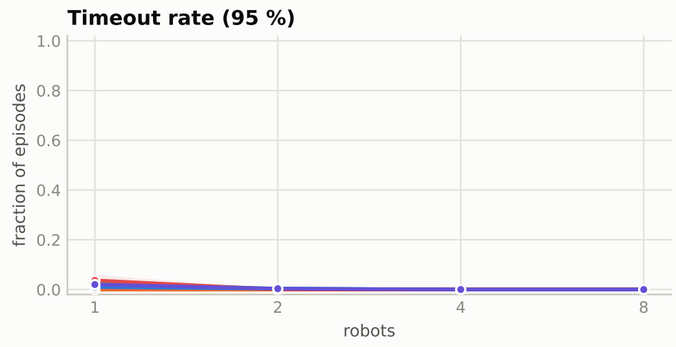
100 maps x 3 episodes, goal 95 % coverage, limit 250 min

Success rate (95 %)

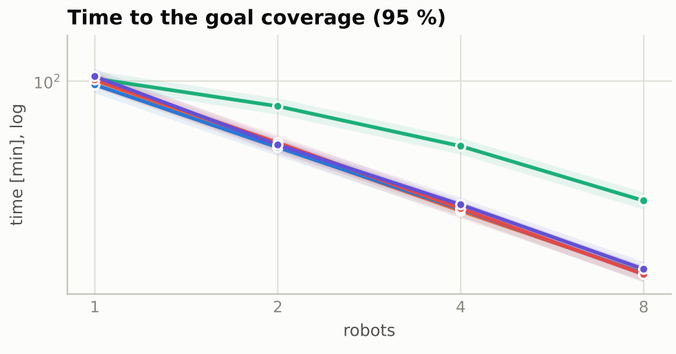


Collision rate

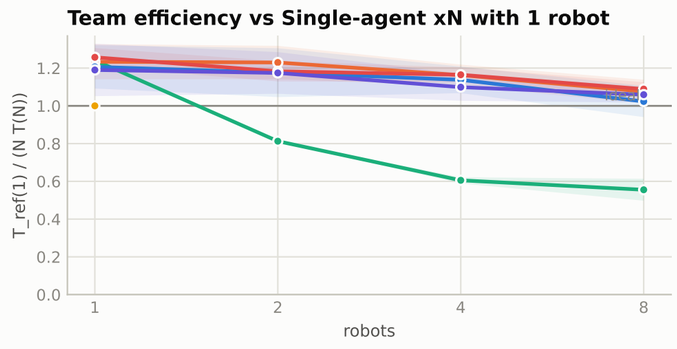
Timeout rate (95 %)



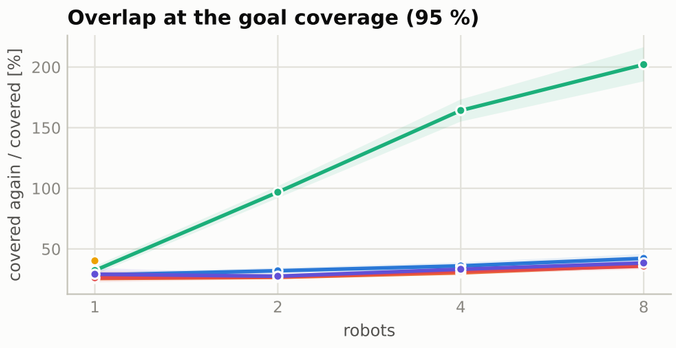
Time to the goal coverage (95 %)



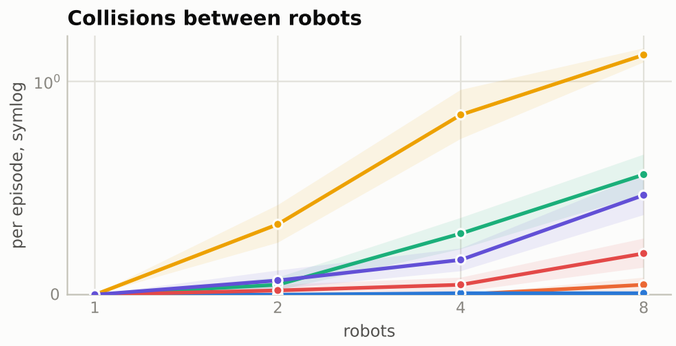
Team efficiency vs Single-agent xN with 1 robot



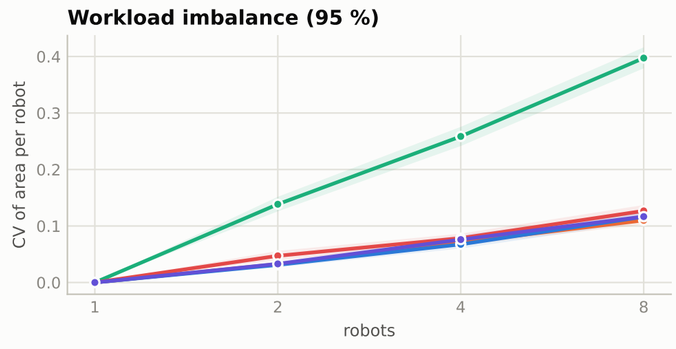
Overlap at the goal coverage (95 %)



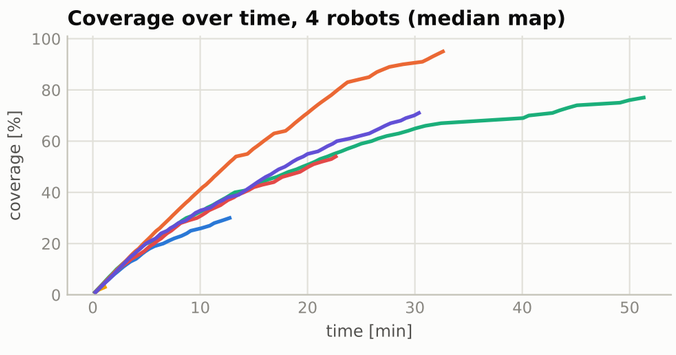
Collisions between robots



Workload imbalance (95 %)



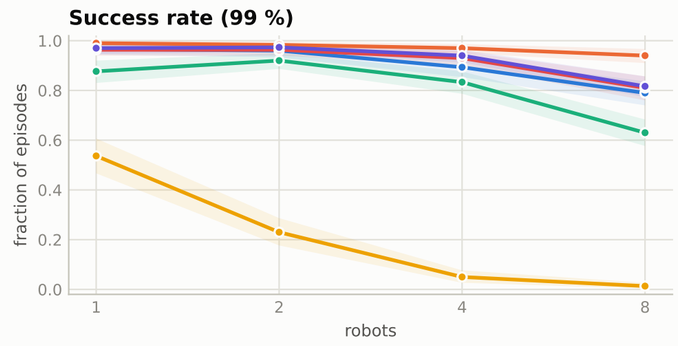
Coverage over time, 4 robots (median map)



test_small

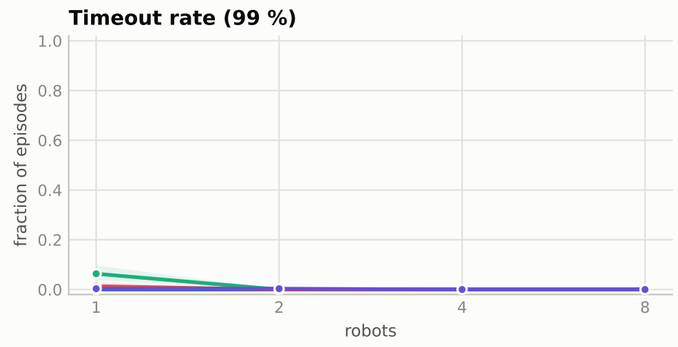
100 maps x 3 episodes, goal 99 % coverage, limit 67 min

Success rate (99 %)

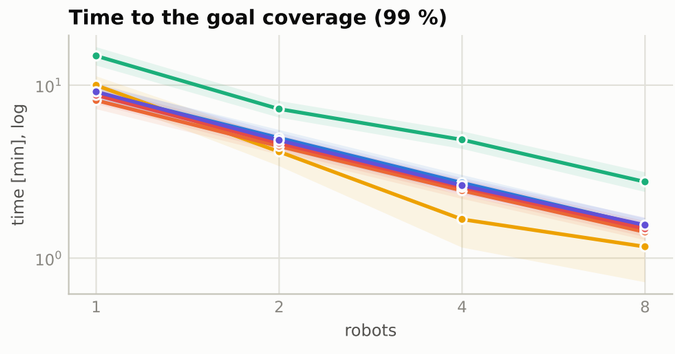


Collision rate

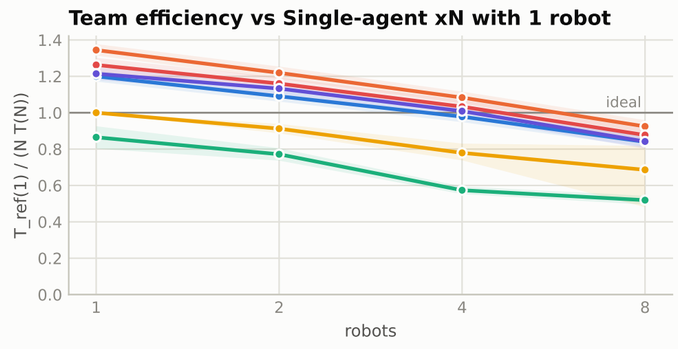
Timeout rate (99 %)



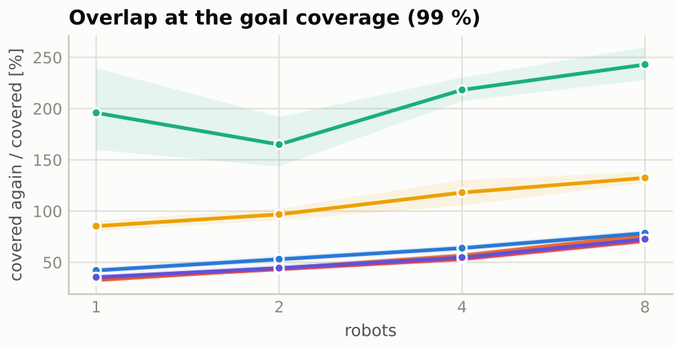
Time to the goal coverage (99 %)



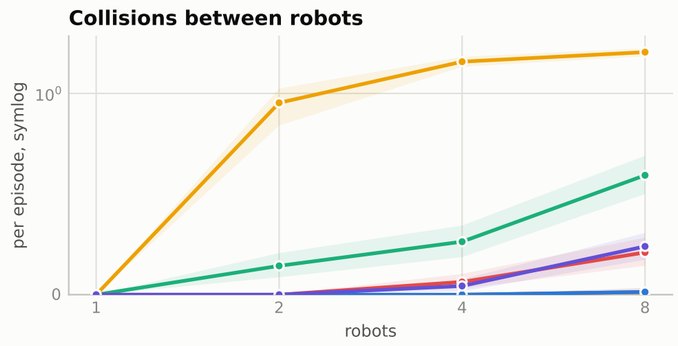
Team efficiency vs Single-agent xN with 1 robot



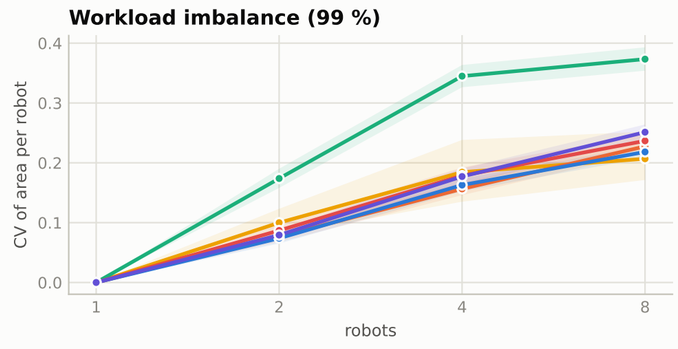
Overlap at the goal coverage (99 %)



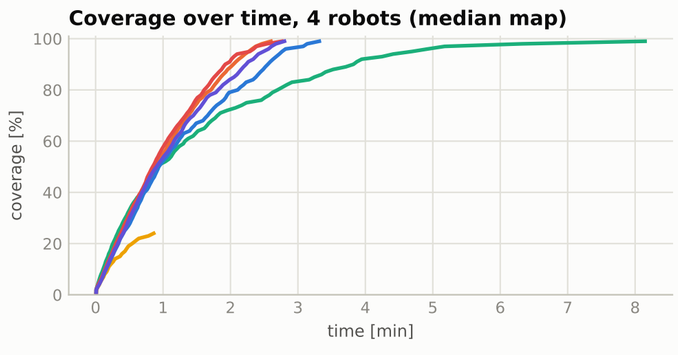
Collisions between robots



Workload imbalance (99 %)



Coverage over time, 4 robots (median map)



Mean over all map sets and team sizes (N = 1, 2, 4, 8), each (map set, N) weighs the same; best of each column highlighted

	Success rate	Collision rate	Timeout rate	Time to goal min	Team efficiency T_ref(1) / (N T(N))	Overlap %	Robot collisions per episode	Imbalance CV of area	Path ratio team path x width / goal area
Single-agent xN	0.089	0.910	**0.000**	**15.0**	0.924	77.4	0.9	0.096	1.89
Single-robot independent xN	0.750	0.234	0.016	44.6	0.761	150.0	0.3	0.219	2.57
IPPO	**0.866**	**0.130**	0.004	32.5	**1.104**	38.0	**0.0**	**0.083**	**1.44**
MAPPO	0.631	0.362	0.006	30.2	1.027	43.8	**0.0**	0.086	1.51
IPPO retrained	0.773	0.221	0.006	32.9	1.064	**37.6**	0.1	0.094	**1.44**
MAPPO retrained	0.745	0.249	0.005	32.8	1.041	38.6	0.1	0.088	1.45